

## CAPE GIRARDEAU, MO, USA

WMO: 723290

Lat: 37.225N

Lon: 89.571W

Elev: 102

StdP: 100.10

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03935

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -12.4      | -9.2 | -18.1                         | 0.8 | -11.3 | -15.2 | 1.0 | -7.8 | 11.8                  | 5.3  | 10.7 | 6.3  | 3.7                   | 350  | 0.468 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.6                   | 34.4            | 25.1 | 33.3 | 24.8 | 32.2 | 24.4 | 26.8                | 32.2 | 26.1 | 31.3 | 25.4 | 30.4 | 3.6                  | 210  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 25.3                            | 20.7 | 30.3 | 24.5 | 19.7 | 29.3 | 23.9 | 19.0 | 28.5 | 84.5          | 31.9 | 81.1 | 31.5 | 78.0 | 30.5 | 30.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 10.0          | 8.7  | 7.8 | -17.6                      | 36.7 | 3.9                | 1.5 | -20.4                                              | 37.8 | -22.7      | 38.7 | -24.9      | 39.5 | -27.7      | 40.6 |    |    |
| (5)               |      |     | -18.1                      | 27.9 | 3.7                | 0.9 | -20.8                                              | 28.6 | -22.9      | 29.1 | -25.0      | 29.5 | -27.7      | 30.2 |    |    |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 14.3   | 1.3   | 3.7   | 8.6   | 14.5  | 19.8  | 24.4  | 25.9  | 25.1  | 21.0  | 14.8  | 8.2   | 3.3  |
|                                                                               | DBStd       | 9.87   | 6.24  | 6.05  | 5.66  | 4.76  | 4.18  | 3.01  | 2.61  | 2.73  | 3.88  | 4.91  | 5.11  | 5.72 |
|                                                                               | HDD10.0     | 899    | 278   | 188   | 94    | 14    | 1     | 0     | 0     | 0     | 12    | 93    | 218   |      |
|                                                                               | HDD18.3     | 2331   | 528   | 408   | 303   | 132   | 35    | 1     | 0     | 1     | 132   | 304   | 466   |      |
|                                                                               | CDD10.0     | 2460   | 8     | 13    | 52    | 148   | 304   | 432   | 494   | 468   | 329   | 161   | 39    | 11   |
|                                                                               | CDD18.3     | 851    | 0     | 0     | 2     | 16    | 81    | 183   | 236   | 211   | 98    | 23    | 1     | 0    |
|                                                                               | CDH23.3     | 8391   | 0     | 0     | 15    | 119   | 669   | 1856  | 2453  | 2100  | 947   | 224   | 8     | 0    |
|                                                                               | CDH26.7     | 3356   | 0     | 0     | 1     | 12    | 190   | 772   | 1072  | 911   | 347   | 52    | 1     | 0    |
| Wind                                                                          | WSAvg       | 3.3    | 4.0   | 4.1   | 4.2   | 4.2   | 3.4   | 2.8   | 2.4   | 2.1   | 2.3   | 2.9   | 3.5   | 3.6  |
| Precipitation                                                                 | PrecAvg     | 1174   | 75    | 82    | 117   | 108   | 133   | 88    | 90    | 91    | 97    | 77    | 106   | 110  |
|                                                                               | PrecMax     | 1735   | 300   | 245   | 302   | 323   | 429   | 282   | 272   | 198   | 260   | 270   | 212   | 313  |
|                                                                               | PrecMin     | 760    | 12    | 23    | 18    | 32    | 41    | 18    | 16    | 8     | 12    | 0     | 9     | 20   |
|                                                                               | PrecStd     | 200    | 58    | 50    | 68    | 64    | 83    | 52    | 58    | 51    | 65    | 55    | 51    | 66   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 18.9  | 21.1  | 25.3  | 28.2  | 32.3  | 35.4  | 36.1  | 36.0  | 33.9  | 30.6  | 24.4  | 19.3 |
|                                                                               |             | MCWB   | 16.5  | 17.2  | 17.6  | 19.8  | 22.8  | 24.2  | 26.3  | 25.4  | 23.7  | 21.9  | 17.3  | 16.8 |
|                                                                               | 2%          | DB     | 16.9  | 18.3  | 22.8  | 26.4  | 30.4  | 33.7  | 34.2  | 34.2  | 32.2  | 28.0  | 21.4  | 17.1 |
|                                                                               |             | MCWB   | 14.9  | 14.4  | 16.7  | 19.1  | 22.1  | 24.2  | 25.8  | 25.2  | 23.4  | 19.9  | 16.4  | 14.7 |
|                                                                               | 5%          | DB     | 14.0  | 16.1  | 20.4  | 24.6  | 28.8  | 32.5  | 33.0  | 32.8  | 30.5  | 25.9  | 19.1  | 14.7 |
|                                                                               |             | MCWB   | 11.6  | 12.7  | 15.1  | 17.9  | 21.3  | 23.8  | 25.5  | 25.0  | 22.8  | 18.6  | 15.3  | 12.4 |
|                                                                               | 10%         | DB     | 11.1  | 13.5  | 18.1  | 22.8  | 27.2  | 31.2  | 31.9  | 31.4  | 28.6  | 23.7  | 17.1  | 12.2 |
|                                                                               |             | MCWB   | 8.8   | 10.1  | 13.6  | 17.2  | 20.8  | 23.4  | 25.0  | 24.3  | 21.5  | 17.7  | 13.7  | 10.1 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 16.9  | 17.9  | 19.4  | 21.8  | 24.6  | 26.4  | 28.0  | 27.8  | 25.6  | 23.2  | 19.6  | 17.3 |
|                                                                               |             | MCDB   | 18.3  | 20.5  | 23.7  | 26.3  | 29.6  | 32.3  | 33.3  | 33.6  | 31.1  | 28.6  | 22.1  | 18.9 |
|                                                                               | 2%          | WB     | 15.2  | 15.5  | 17.7  | 20.4  | 23.4  | 25.6  | 27.1  | 26.7  | 24.6  | 21.7  | 17.8  | 15.1 |
|                                                                               |             | MCDB   | 16.7  | 17.8  | 21.7  | 24.5  | 28.4  | 31.5  | 32.2  | 32.2  | 30.1  | 26.1  | 20.1  | 16.8 |
|                                                                               | 5%          | WB     | 12.3  | 13.4  | 16.0  | 19.2  | 22.5  | 24.9  | 26.4  | 25.8  | 23.7  | 20.3  | 16.0  | 12.8 |
|                                                                               |             | MCDB   | 13.5  | 15.5  | 19.3  | 23.1  | 27.2  | 30.6  | 31.6  | 31.0  | 28.6  | 23.8  | 18.6  | 14.3 |
|                                                                               | 10%         | WB     | 8.8   | 10.3  | 14.3  | 18.0  | 21.6  | 24.2  | 25.6  | 25.1  | 22.8  | 18.8  | 13.9  | 10.2 |
|                                                                               |             | MCDB   | 10.8  | 12.8  | 17.6  | 21.7  | 26.0  | 29.4  | 30.5  | 30.0  | 26.9  | 22.5  | 16.6  | 12.0 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 9.1   | 9.8   | 11.0  | 11.8  | 10.9  | 11.1  | 10.6  | 11.7  | 13.1  | 13.1  | 10.8  | 8.8  |
|                                                                               |             | MCDBR  | 11.5  | 12.8  | 13.4  | 12.9  | 12.4  | 12.5  | 11.5  | 13.0  | 14.3  | 14.8  | 13.5  | 10.8 |
|                                                                               |             | MCWBR  | 9.7   | 9.9   | 8.8   | 7.2   | 5.8   | 4.7   | 4.7   | 5.6   | 6.5   | 7.6   | 9.4   | 9.0  |
|                                                                               | 5% WB       | MCDBR  | 9.8   | 11.4  | 11.7  | 10.7  | 10.5  | 10.8  | 10.4  | 11.3  | 12.0  | 11.9  | 11.3  | 9.8  |
|                                                                               |             | MCWBR  | 9.3   | 9.9   | 8.6   | 7.1   | 5.8   | 4.9   | 4.8   | 5.4   | 5.9   | 6.8   | 9.3   | 9.0  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.315  | 0.328 | 0.361 | 0.395 | 0.431 | 0.443 | 0.481 | 0.476 | 0.411 | 0.356 | 0.322 | 0.308 |      |
|                                                                               | taud        | 2.470  | 2.416 | 2.384 | 2.286 | 2.243 | 2.223 | 2.145 | 2.152 | 2.319 | 2.460 | 2.500 | 2.494 |      |
|                                                                               | Ebn at Noon | 873    | 904   | 902   | 885   | 855   | 842   | 807   | 802   | 839   | 861   | 859   | 857   |      |
|                                                                               | Edh at Noon | 83     | 98    | 112   | 130   | 138   | 141   | 152   | 147   | 118   | 93    | 80    | 76    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 2.19   | 2.80  | 3.85  | 5.05  | 5.70  | 6.46  | 6.23  | 5.70  | 4.77  | 3.59  | 2.45  | 1.85  |      |
|                                                                               | RadStd      | 0.22   | 0.35  | 0.34  | 0.34  | 0.47  | 0.40  | 0.37  | 0.31  | 0.28  | 0.39  | 0.27  | 0.17  |      |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (27 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | +73     |

Nomenclature: See separate page